Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

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Advanced Algorithms and Parallel Programming – II part – September 11th, 2023

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about Parallel Algorithms and Parallel Programming and briefly explain (without an explanation, the answer will be considered invalid)

A. Could you please describe the statement/operation dependency and how and when it can be removed? (2)

B. Under which condition the following code is parallelizable? In case the conditions hold, are there better parallelizable solutions? (2)

```
for (i=0; i<100; i++)
        for (j=0; j<10; j++)
            a[i][j] = f(a[i][j]);
```

Exercise n. 2

Answer the following questions about parallel patterns and briefly explain (without an explanation, the answer will be considered invalid)

A. Please describe the reduce pattern: assumptions, complexity, and example. (2)

B. Please describe how the Unpack parallel pattern could be implemented. (2)

Exercise n. 3

Answer the following questions about OpenMP and briefly explain (without an explanation, the answer will be considered invalid)

A. Please describe what is meant by task parallelism (Parallel Task Graph) and how could be implemented in OpenMP?(2)

B. Please describe the simd directive in OpenMP. (2)

Exercise n. 4

Answer the following questions about MPI and domain-specific languages and briefly explain (without explanation, the answer will be considered invalid)

A. Describe the functionality and the parameters meaning of the function MPI_Gather(You can skip this question if you completed the second challenge) (2)

B. Please briefly discuss what the term "schedule" means in the domain-specific language Halide. (2)